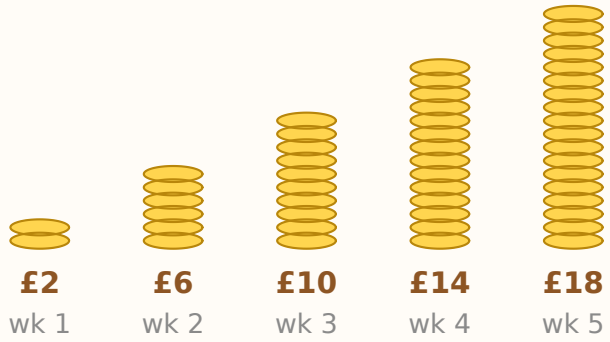
Chapter 17

Sequences and series

patterns that march on, and adding them all up



Each week you save £4 more than the week before. How much after a year?

$$2 + 6 + 10 + \cdots = ?$$

A **sequence** is a list of numbers that follows a rule, like the savings above: 2, 6, 10, 14, ... A **series** is what you get when you **add** the list up. This chapter has two kinds: lists that go up by the **same amount** each time (arithmetic) and lists that get **multiplied** by the same amount each time (geometric).

For each kind you'll learn to find any term without writing out the whole list, and to add up hundreds of terms in one line. Along the way, a trick a ten-year-old found, a Python loop, a bouncing ball, and an endless sum that adds up to exactly 1.

Goes with **Lecture 13** (arithmetic and geometric progressions, summation notation).

Terms live in numbered lockers

Each number in a sequence is a **term**. Terms have positions: 1st, 2nd, 3rd... Picture a row of lockers, each with a number on the door (the position n) and a value inside:

n = 1	n = 2	n = 3	n = 4	n = 5	n = 6	...
2	6	10	14	18	22	

The value in locker n is often written u_n ("u sub n"). Here $u_1 = 2$, $u_4 = 14$. **Sequence** = the lockers. **Series** = the total of what's inside them: $2 + 6 + 10 + \ldots$

Lecture 13 says "series are lists". Strictly, a list is a sequence and a sum is a series; the slides use the words loosely, so don't let that throw you.

Arithmetic progressions: a staircase of equal steps

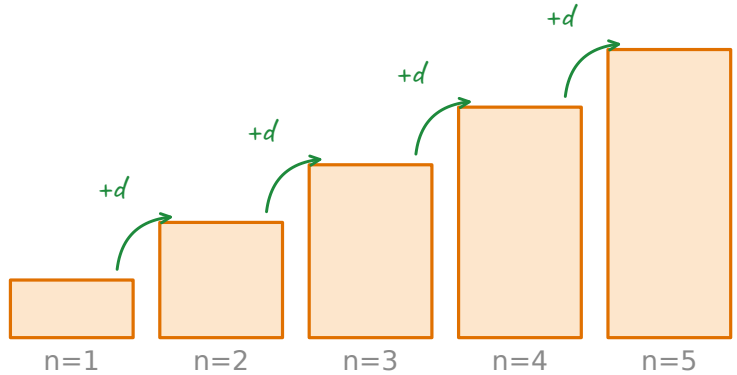
RULE: arithmetic progression (A.P.)

Each term = the previous term **+ d**. The common difference is $d = u_2 - u_1$ (any term minus the one before). a is the first term.

nth term: $u_n = a + (n-1)d$ (to reach term n you make $n-1$ jumps of d).

Any one of u_n, a, n, d can be found from the other three by rearranging.

When each term is the last one **plus a fixed amount**, the sequence is an **arithmetic progression** (A.P.). The fixed amount is the **common difference** d. The savings: $a = 2$ (first term), $d = 4$.



To reach step 5 you make only 4 jumps of d. (Like fence posts: 5 posts, 4 gaps between them.)

So the nth term is the first term plus $(n-1)$ jumps:

$$u_n = a + (n-1)d$$

$$u_8 = 2 + (8-1) \times 4 = 2 + 28 = 30$$ ← 8th week's saving: 7 jumps of 4

Working backwards. The same formula rearranges (Chapter 14) to answer the other questions Lecture 13 asks:

$$n = \frac{l-a}{d} + 1$$ ← how many terms, if the last term is l

$$d = \frac{l-a}{n-1}$$ ← the difference, if you know n

How many terms in 2, 6, ..., 50? The gap from 2 to 50 is 48, that's $48 \div 4 = 12$ jumps, so **13** terms (one more post than gaps). And the slide's check: 2, ..., 30 with 8 terms has $d = \frac{30-2}{8-1} = \frac{28}{7} = 4$ ✓.

Adding them up: Gauss's pairing trick

RULE: sum of an arithmetic progression

First n terms, when you know the last term l: $S_n = \frac{n}{2}(a + l)$.

When you know d instead: $S_n = \frac{n}{2}\big(2a + (n-1)d\big)$.

Both say: (average of the first and last term) × (number of terms). The reason is below.

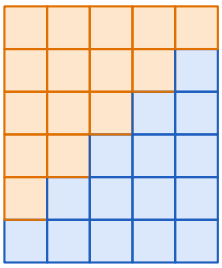
A famous story: a teacher told a class to add 1 + 2 + ... + 100, hoping for a quiet hour. Young Carl Gauss answered in seconds. He wrote the list forwards and backwards and added the columns:

1	2	3	…	99	100
100	99	98	…	2	1
101	101	101	…	101	101

every column makes 101!

100 columns of 101 is 10,100. But that's the list added **twice** (once each way), so the answer is half: **5050**.

The picture: a staircase of blocks, plus the same staircase upside down, makes a rectangle:



5 columns, each 6 tall

blue: 1 + 2 + 3 + 4 + 5 (the staircase)
orange: the same staircase, flipped
together: a 5 × 6 rectangle = 30 blocks
so the staircase alone is 30 ÷ 2 = 15

In general: **(first + last) is the same for every pair, there are n terms, and we halve**. Or in words: the **average term** (halfway between first and last) **times how many terms**:

$$S_n = \frac{a + l}{2} \times n$$

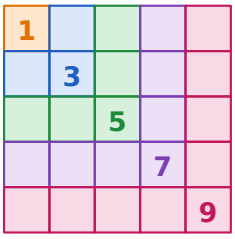
$$2 + 6 + 10 + 14 = \frac{2 + 14}{2} \times 4 = 32$$ ← *Lecture 13, example 1*

$$2 + 6 + \cdots + 50 = \frac{2 + 50}{2} \times 13 = 338$$ ← example 2 (13 terms, from the last page)

If you know n but not the last term, swap l for $a + (n-1)d$: $S_n = \frac{n}{2}\big(2a + (n-1)d\big)$. Slide example: 2 + 6 + 10 + …, 8 terms: $\frac{8}{2}(4 + 7 \times 4) = 4 \times 32 = 128$.

The piggy bank, answered: 52 weeks of 2, 6, 10, … is $\frac{52}{2}(4 + 51 \times 4) = 26 \times 208 = 5408$, so £5408.

A special staircase: the odd numbers. Add $1 + 3 + 5 + 7 + 9$. Each odd number is an L-shaped layer that wraps around the previous square and makes the next one:



each colour is one layer; its number sits in its corner

1 + 3 + 5 + 7 + 9 = 5 × 5

Each new layer adds one row and one column to the square, plus the shared corner: k + k + 1 = the next odd number.

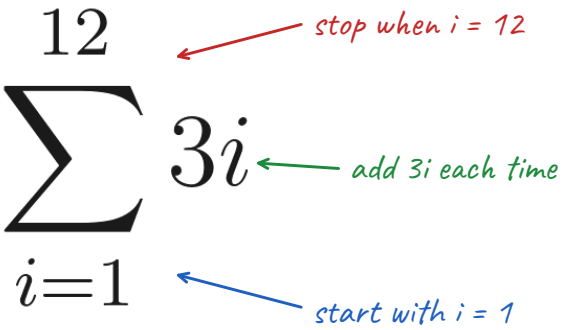
So the first n odd numbers always add up to n^2.

$$S_n = \frac{n}{2}\big(2 \times 1 + (n-1) \times 2\big) = \frac{n}{2} \times 2n = n^2$$

the A.P. formula agrees (a = 1, d = 2)

Σ is a for-loop

Writing $3 + 6 + 9 + \cdots + 36$ in full is tedious, so Lecture 13 uses the Greek capital **sigma**, Σ, meaning "add up". If you've written any code, it's a loop:




```
total = 0
for i in range(1, 13):
    total += 3 * i
print(total)   # 234
```

Read it as: "for i from 1 to 12, add $3i$": $3(1) + 3(2) + \cdots + 3(12)$. The letter i is just a counter; it disappears from the answer. Python's range stops **before** its second number, which is why the code says 13.

Lecture 13 writes the same list a second way, counting from 0: $\sum_{i=0}^{11}(3 + 3i) = 3 + 6 + \cdots + 36$. Different counter, same terms, same total. (It's an A.P., so $\frac{3+36}{2} \times 12 = 234$ ✓.)

Geometric progressions: a bouncing ball

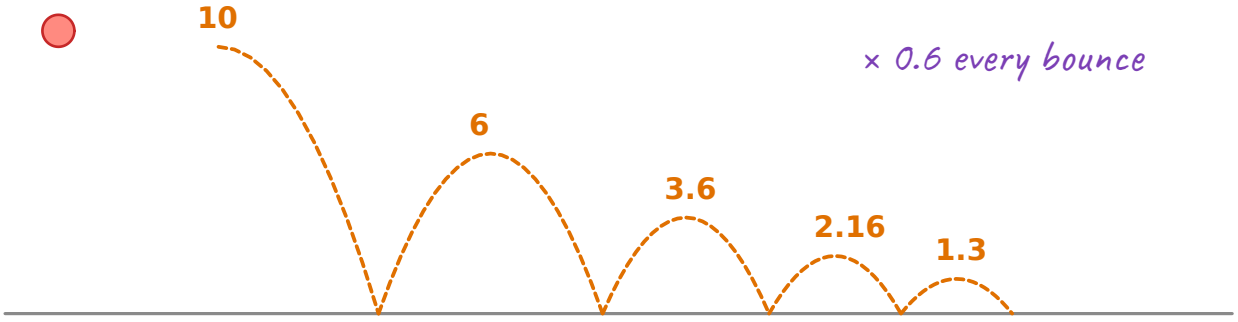
RULE: geometric progression (G.P.)

Each term = the previous term **× r**. The common ratio is $r = \dfrac{u_2}{u_1}$ (any term divided by the one before). a is the first term.

nth term: $u_n = ar^{n-1}$ ($n-1$ multiplications by r).

r can be a fraction (the terms shrink) or negative (the signs alternate).

Drop a ball from 10 m. Each bounce reaches 60% of the height before. The heights are 10, 6, 3.6, 2.16, ... Each term is the last one **multiplied** by the same number, 0.6:



That's a **geometric progression** (G.P.). The multiplier is the **common ratio** r. Find it by dividing any term by the one before: $6 \div 10 = 0.6$.

Term n has been multiplied by r exactly $(n-1)$ times (fence posts again):

$$u_n = ar^{n-1}$$

Lecture 13's example: 3, 6, 12, ..., 3072. Here $a = 3$, $r = 2$. Which term is 3072?

$$3 \times 2^{n-1} = 3072$$

$$2^{n-1} = 1024 = 2^{10}$$ ← ÷3; then keep doubling 1 until 1024: 10 doublings

$$n - 1 = 10 \Rightarrow n = 11$$ ← so 3072 is the 11th term

(The slide's footnote says 3072 is 3×2^{11}; that's 6144. It's 3×2^{10}, and the slide itself then correctly uses $n = 11$.)

Adding a G.P.: shift and subtract

RULE: sum of a geometric progression

First n terms: $S_n = \dfrac{a(1-r^n)}{1-r}$, or the same thing as $\dfrac{a(r^n-1)}{r-1}$ (tidier when $r > 1$).

Only for $r \neq 1$ (otherwise you divide by 0). If $r = 1$ every term is a, so $S_n = na$.

Pairing doesn't work here (the pairs aren't equal). Instead, use this trick: write the sum, then write it again **multiplied by** r. Almost everything lines up:

$$\begin{aligned} S &= a + ar + ar^2 + \ldots + ar^{n-1} \\ rS &= \quad\; ar + ar^2 + \ldots + ar^{n-1} + ar^n \end{aligned}$$

the blue parts match: subtracting wipes them out

$$S - rS = a - ar^n \quad \longleftarrow \text{top line minus bottom line}$$

$$S(1-r) = a(1-r^n) \quad \longleftarrow \text{factorise both sides (Chapter 12)}$$

$$S_n = \frac{a(1-r^n)}{1-r} \quad \longleftarrow \div (1-r)$$

Lecture 13: $3 + 6 + 12 + \cdots + 3072$, with $a = 3$, $r = 2$, $n = 11$:

$$S_{11} = \frac{3(1-2^{11})}{1-2} = \frac{3(1-2048)}{-1} = 6141$$

(The slide puts a $\sum$ in front of this formula. It doesn't need one: the formula already **is** the total.)

When $r = 1$ the formula gives $\frac{a \times 0}{0}$, which means nothing. But then the list is just $a, a, a, \ldots$, so add it directly: $4 + 4 + 4 + \ldots$ (10 terms) $= 10 \times 4 = 40$.

Negative ratio: $2 - 6 + 18 - 54 + 162$ has $a = 2$, $r = -3$, $n = 5$:

$$S_5 = \frac{2(1-(-3)^5)}{1-(-3)} = \frac{2(1+243)}{4} = 122$$

$(-3)^5 = -243$: keep the brackets

Walking halfway to the wall, forever

RULE: sum to infinity of a G.P.

$S_\infty = \frac{a}{1-r}$, **only when** $-1 < r < 1$ (written $|r| < 1$). Then $r^n \to 0$ and the sums settle on this value: the series **converges**.

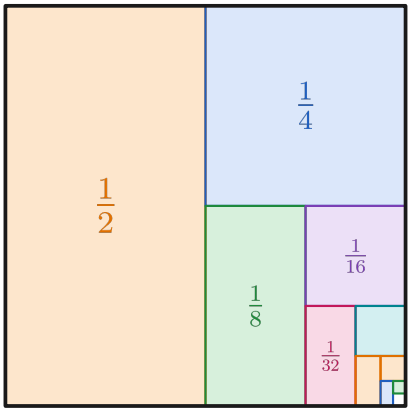
No sum to infinity when $r \geq 1$ or $r \leq -1$: the terms don't shrink, so the total grows without limit or swings about forever.

Stand 1 m from a wall. Walk half the distance, then half what's left, then half again... You walk $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots$ and never stop, yet you never pass the wall:



the pieces fill the gap closer and closer: the total heads to exactly 1 m

The same sum, folded into a square. Let a square have area 1. Shade half of it, then half of what's left, then half of that...



After 1/2 + 1/4 + 1/8 the white part is 1/8.
After 10 pieces it's 1/1024: a speck.

The pieces never spill outside the square, and the white gap halves every time. So the total can't be more than 1, and it gets as close to 1 as you like: it IS 1.

When the terms shrink (the ratio is between −1 and 1), r^n gets closer and closer to 0 as n grows ($0.5^{10} \approx 0.001$). So in the G.P. sum formula the r^n fades away:

$$S_\infty = \frac{a(1-0)}{1-r} = \frac{a}{1-r}$$ ← valid only if $-1 < r < 1$

The slides write the condition as $|r| < 1$: the bars mean "size, ignoring the sign" (the **modulus**). Wall walk: $a = \frac{1}{2}$, $r = \frac{1}{2}$: $\frac{1/2}{1-1/2} = 1$ ✓.

Slide check: Lecture 13 writes $S_\infty = \frac{a}{r-1}$. For the wall walk that gives $\frac{1/2}{-1/2} = -1$: adding up positive distances can't give a negative total! The bottom must be $1 - r$.

The ball, answered: it falls 10 m, then goes up and down each bounce height: $10 + 2(6 + 3.6 + 2.16 + \dots) = 10 + 2 \times \frac{6}{1-0.6} = 10 + 30 = 40$ m in total.

When there is no sum to infinity. If $r \geq 1$ (e.g. 3, 6, 12, ...) the total just keeps growing. If $r \leq -1$ the terms never shrink either: with $r = -1$, $1 - 1 + 1 - 1 + \dots$ flips between 1 and 0 forever; with $r = -2$, $1 - 2 + 4 - 8 + \dots$ swings wider and wider. Either way it never settles, so there is no S_∞. **Always check $-1 < r < 1$ before using $\frac{a}{1-r}$**: blindly using it for $r = 2$ gives $\frac{1}{1-2} = -1$ for $1 + 2 + 4 + \dots$, which is nonsense.

Which kind is it? The gaps test and the ratio test

Given a list, check the **gaps** (subtract neighbours) and the **ratios** (divide neighbours). Whichever stays constant tells you the type:

list	gaps	ratios	verdict
5, 8, 11, 14	3, 3, 3	1.6, 1.375, ...	**A.P., d = 3**
2, 10, 50, 250	8, 40, 200	5, 5, 5	**G.P., r = 5**
80, 40, 20, 10	−40, −20, −10	0.5, 0.5, 0.5	**G.P., r = 0.5**
1, 4, 9, 16	3, 5, 7	4, 2.25, ...	**neither (squares)**

A.P. and G.P. rules side by side

Every formula in this chapter, in one table. The left column adds the same amount each time; the right column multiplies by the same amount.

	arithmetic (A.P.)	geometric (G.P.)
each step	add d	multiply by r
find the step	$d = u_2 - u_1$	$r = \dfrac{u_2}{u_1}$
nth term	$u_n = a + (n-1)d$	$u_n = ar^{n-1}$
sum of n terms	$S_n = \dfrac{n}{2}\big(2a + (n-1)d\big)$ or $S_n = \dfrac{n}{2}(a + l)$	$S_n = \dfrac{a(1-r^n)}{1-r}$ ($r \neq 1$; $r = 1$: $S_n = na$)
sum to infinity	none: the terms don't shrink to 0, so the sum has no limit	$S_\infty = \dfrac{a}{1-r}$ **only if** $-1 < r < 1$
three terms in a row x, y, z	$y - x = z - y$	$\dfrac{y}{x} = \dfrac{z}{y}$

Find the unknown: working backwards

PROCEDURE: given some facts, find the rest

1. Write each fact as an equation using the formulas: "the 3rd term is 11" means $a + 2d = 11$; "the sum is 210" means $S_n = 210$.
2. Two unknowns (a and d, or a and r) need two equations. For an A.P. subtract them; for a G.P. **divide** them, so a cancels.
3. Solve, then check by writing out the first few terms. Reject answers that make no sense (a negative or fractional number of terms).

Example: an A.P. has 3rd term 11 and 8th term 31. Find a and d.

$$a + 2d = 11, \qquad a + 7d = 31$$ ← u_3 and u_8 as equations

$$5d = 20 \;\Rightarrow\; d = 4$$ ← subtract: a cancels

$$a + 2(4) = 11 \;\Rightarrow\; a = 3$$ ← back into the first

Check: 3, 7, **11**, 15, 19, 23, 27, **31** ✓

Example: how many terms of $3 + 7 + 11 + \ldots$ add up to 210?

$$\frac{n}{2}\big(2(3) + (n-1)4\big) = 210$$ ← A.P. sum with a = 3, d = 4

$$\frac{n}{2}(4n + 2) = n(2n + 1) = 210 \;\Rightarrow\; 2n^2 + n - 210 = 0$$

a quadratic in n

$$(2n + 21)(n - 10) = 0 \;\Rightarrow\; n = 10$$ ← n = −10.5 is rejected: n counts terms

Example: a G.P. has 2nd term 6 and 5th term 162. Find a and r.

$$ar = 6, \qquad ar^4 = 162$$ ← u_2 and u_5 as equations

$$\frac{ar^4}{ar} = r^3 = 27 \;\Rightarrow\; r = 3$$ ← divide: a cancels

$$a = \frac{6}{3} = 2$$

Check: 2, **6**, 18, 54, **162** ✓

Example: a G.P. has first term 5 and sum to infinity 20. Find r.

$$\frac{5}{1 - r} = 20 \;\Rightarrow\; 1 - r = \tfrac{1}{4} \;\Rightarrow\; r = \tfrac{3}{4}$$ ← $S_\infty = a/(1 - r)$

$r = \frac{3}{4}$ is between −1 and 1, so a sum to infinity really exists ✓.

Example: what is the smallest number of terms of $1 + 2 + 4 + 8 + \ldots$ that add to more than 1000?

$$S_n = \frac{1(2^n - 1)}{2 - 1} = 2^n - 1 > 1000 \;\Rightarrow\; 2^n > 1001$$

a = 1, r = 2

$$2^9 = 512, \quad 2^{10} = 1024 \;\Rightarrow\; n = 10$$ ← try powers of 2

Trial works because the powers are easy. Chapter 22 solves $2^n > 1001$ directly with logs.

Example: k, $2k+1$ and $4k-2$ are consecutive terms of an A.P. Find k.

$$(2k+1) - k = (4k-2) - (2k+1) \quad \leftarrow \text{equal gaps}$$

$$k + 1 = 2k - 3 \;\Rightarrow\; k = 4 \quad \leftarrow \text{terms 4, 9, 14 ✓}$$

Mistakes to avoid

✗ Common mistakes

✗ $u_n = a + nd$

n − 1 jumps to reach term n

✓ $u_n = a + (n-1)d$

✗ $u_n = ar^n$

the first term has had no multiplications

✓ $u_n = ar^{n-1}$

✗ $r = 2: \; S_\infty = \dfrac{a}{1-2}$

only when −1 < r < 1

✓ $r = 2: \;$ no S_∞

✗ $S_\infty = \dfrac{a}{r-1}$

1 minus r on the bottom

✓ $S_\infty = \dfrac{a}{1-r}$

✗ $-3^5 = 243$

brackets round a negative ratio

✓ $(-3)^5 = -243$

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

A.P. term	$u_n = a + (n-1)d$ d = any term minus the one before
A.P. sum	$S_n = \frac{n}{2}\big(2a + (n-1)d\big)$
A.P. sum (last term l)	$S_n = \frac{(a+l)}{2}\,n$
Number of terms	$n = \frac{l-a}{d} + 1$ one more term than gaps
G.P. term	$u_n = ar^{n-1}$ r = any term divided by the one before
G.P. sum	$S_n = \frac{a(1-r^n)}{1-r}$ $r \neq 1$; if $r = 1$, $S_n = na$
Sum to infinity	$S_\infty = \frac{a}{1-r} \quad (\lvert r\rvert < 1)$ none if $r \geq 1$ or $r \leq -1$
Sigma	$\sum_{i=1}^{n} f(i) = f(1) + f(2) + \cdots + f(n)$ a for-loop: add for each counter value
Odd numbers	$1 + 3 + 5 + \cdots + (2n-1) = n^2$
Two terms given	two equations; subtract (A.P.) or divide (G.P.)
Sum given, find n	set S_n equal to the total and solve for n (often a quadratic); n must be a positive whole number

Piggy banks and bouncing balls: your turn

Answers in grey. For sums, try the first few terms by hand to check your formula.

Arithmetic

1. Find the 20th term of 7, 10, 13, ...
2. How many terms in 5, 9, 13, ..., 81?
3. Add 1 + 2 + ... + 50
4. Add 7 + 10 + 13 + ... (20 terms)
5. An A.P. starts at 4 and its 10th term is 31. Find d.

1. 64 2. 20 3. 1275 4. 710 5. 3

Sigma

6. Write out and add: Σ from i = 1 to 5 of (2i + 1)
7. Write 4 + 8 + 12 + ... + 40 using Σ

6. 3 + 5 + 7 + 9 + 11 = 35 7. Σ from i = 1 to 10 of 4i (total 220)

Geometric

8. Find the 8th term of 2, 6, 18, ...
9. Add 5 + 10 + 20 + ... (8 terms)
10. Sum to infinity: 12 + 6 + 3 + ...
11. Sum to infinity: 1 − 1/3 + 1/9 − ...
12. Which is 1, 1.5, 2.25, 3.375: A.P. or G.P.? Does it have a sum to infinity?

8. 4374 9. 1275 10. 24 11. 3/4 12. G.P. with r = 1.5; no (r > 1)

Find the unknown

13. An A.P. has 4th term 17 and 10th term 41. Find a and d.
14. How many terms of 4 + 7 + 10 + ... add up to 246?
15. A G.P. has 3rd term 12 and 6th term 96. Find a and r.
16. A G.P. has first term 8 and sum to infinity 12. Find r.
17. Smallest n for which 3 + 6 + 12 + ... (n terms) is more than 1000?
18. x, x + 3 and x + 9 are consecutive terms of a G.P. Find x.

13. d = 4, a = 5 (6d = 24) 14. n = 12 ($3n^2 + 5n - 492 = 0$; reject the negative root) 15. r = 2, a = 3 ($r^3 = 96/12 = 8$) 16. r = 1/3 ($8/(1 - r) = 12$) 17. n = 9 ($S_8 = 3(2^8 - 1) = 765$, $S_9 = 1533$) 18. x = 3 ($(x + 3)^2 = x(x + 9)$, so 3x = 9; terms 3, 6, 12)